

FIR Management System - Project Documentation

1. Project Overview

The FIR Management System is a web-based application designed for managing and reporting First Information Reports (FIRs). It features a robust backend using Node.js and PostgreSQL (Supabase), and a modern, responsive frontend with Progressive Web App (PWA) capabilities.

2. Implementation History

Phase 1: CSV Data Integration (Implemented & Reverted)

- **Objective**: Import bulk data from ``data.csv``.
- **Implementation**:
 - Added a POST route ``/admin/import`` in ``app.js``.
 - Used ``csv-parser`` to process records.
 - Normalized Police Station names (e.g., "Extention" to "Extension").
 - Formatted dates from ``M/D/YY`` to ``YYYY-MM-DD``.
- **Reversion**: Per user request, the import feature and database connection strings were reverted to their original state to maintain codebase simplicity.

Phase 2: Reporting & Sorting Improvements

- **Objective**: Improve the usability of FIR reports.
- **Changes**:
 - **Natural Sorting**: Modified the sorting logic to ensure FIR numbers (e.g., "1/2024", "10/2024") are sorted in **descending order** (highest numbers first) globally.
 - **Export Symbol Fixes**: Cleaned the "Status" column in PDF and Excel exports by removing emojis (``'``, ``&``, ``#ñ``) that caused encoding errors (``#ñ``).

Phase 3: Premium UI Overhaul (Login Page)

- **Objective**: Create a high-quality, professional first impression.
 - **Changes**:
 - **Visual Branding**: Added floating police logos flanking the login box.
 - **Developer Credit**: Added a subtle "G-NTTF" developer logo in the footer.
 - **Mobile Optimization**: Reconfigured the layout for mobile devices to stack logos and maintain a clean form factor.
 - **Cache Busting**: Implemented versioning (``?v=1``) on assets to ensure updates reflect immediately across all devices.
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3. Technical Configuration

Environment Variables (.env)

- ``DATABASE_URL``: Connected to Supabase PostgreSQL.

- `PORT`: Defaulting to 3000.

Key Dependencies

- `express`: Web framework.
 - `pg`: PostgreSQL client.
 - `ejs`: Templating engine.
 - `xlsx`: Excel export handling.
 - `jspdf`: PDF export handling.
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4. Maintenance & Support

- **PWA Support**: The application includes a service worker for offline support and manifest for "Add to Home Screen" functionality.
 - **Auto-Update**: The login page includes a script to force refresh when a new version is detected by the service worker.
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